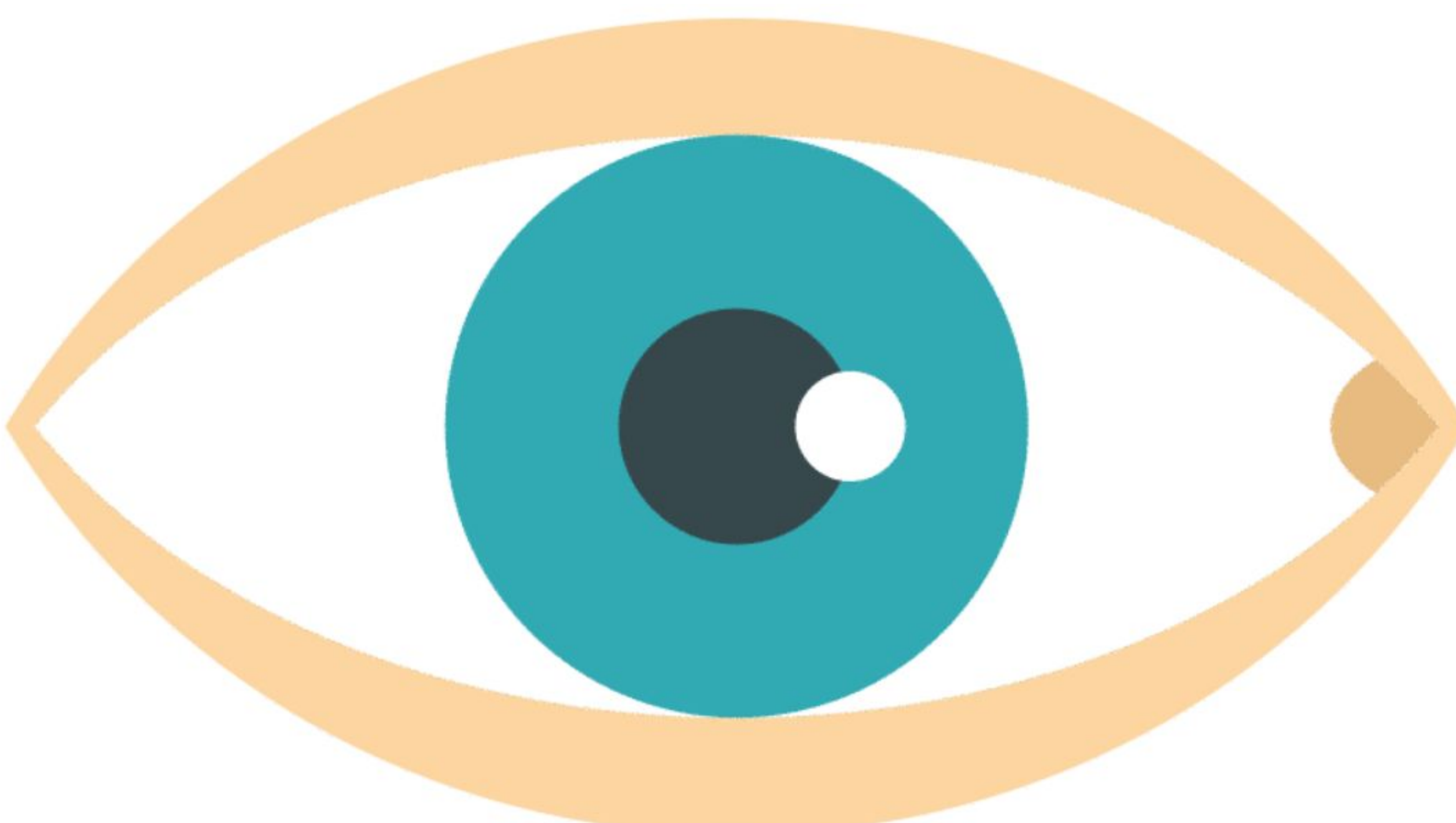
PLABABLE GEMS

VERSION 1.1

OPHTHALMOLOGY



Conjunctivitis

Bacterial	Viral	Allergic
Purulent sticky discharge	-Serous discharge -Preauricular -Lymphadenopath y	-Bilateral red eye -Serous discharge and itching
H/o eyes sticking together after waking up	Recent h/o URTI Common: **Adenovirus**	H/o atopy and seasonal variation
Rx: Symptomatic till 2 weeks **Topical antibiotic** - chloramphenicol after 2 weeks or if the condition is severe	**Rx:** Symptomatic (cold presses and artificial tears)	**Rx:** Topical mast cell stabilizer and antihistamines Avoidance of allergen

Conjunctivitis

Bacterial conjunctivitis

Viral conjunctivitis

Ophthalmia Neonatorum

Conjunctivitis in a newborn within 28 days of birth

Most common organism involved is *Chlamydia trachomatis* followed by *N. gonorrhoeae*

Presentation

- Purulent or mucopurulent discharge
- Injected conjunctiva
- Lid swelling

Treatment

- **Chlamydial infection:** Oral erythromycin for 14 days or azithromycin for 3 days
- **Gonorrhoeal infection:** Ceftriaxone IV or IM single dose
- **Viral:** IV acyclovir for 14 days

Prevention: Screening of mother for STI

Anterior uveitis

Inflammation of the iris, also known as iritis

Presentation

- Pain, redness and photophobia
- Usually unilateral
- Blurring of vision
- Pupil may be abnormally shaped due to iris spasm
- **Sign** - Presence of cells in the anterior chamber

Associated conditions

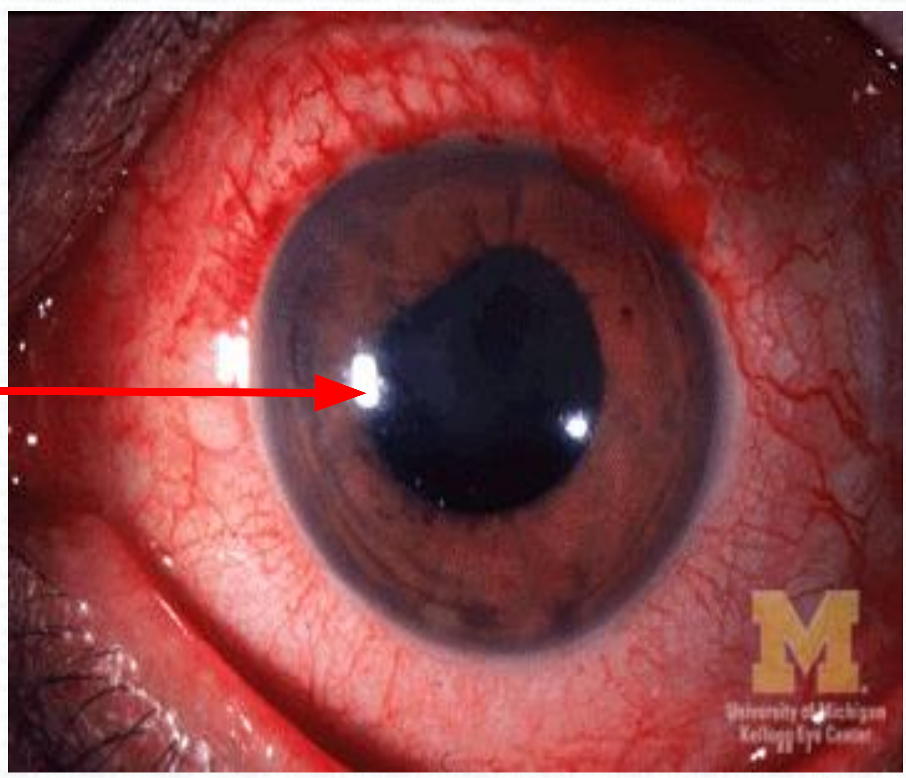
- Seronegative arthropathies -Ankylosing spondylitis and reactive arthritis (HLA B27)
- Inflammatory bowel diseases
- Sarcoidosis
- Behcet's disease

Treatment

- Corticosteroids
- Cyclopentolate to relieve pain and prevent adhesions
- Ciclosporin if recurrent and affects visual acuity

Anterior uveitis

Irregularly shaped pupil



Thyroid Eye Disease

Brain trainer:

A female patient presents with diplopia, tachycardia, lid lag and restricted eye movements. Which investigation should you order?

➔ **Thyroid function test**

Acute Dacryocystitis

Brain trainer:

A patient presents with excess tears, pain, redness and swelling in the left eye near the region of the lacrimal sac. What is the most likely diagnosis?

➔ **Acute dacryocystitis**

Cytomegalovirus Retinitis

Brain trainer:

A HIV attends with visual deterioration. What should be on the top of your differential?

➔ **Cytomegalovirus retinitis**

Chalazion

Brain trainer:

A woman with a lump over the lower eyelid and without any other findings / complaints. What is the diagnosis and management?

- ➔ **Meibomian cyst (chalazion)**
- ➔ **Warm compresses**

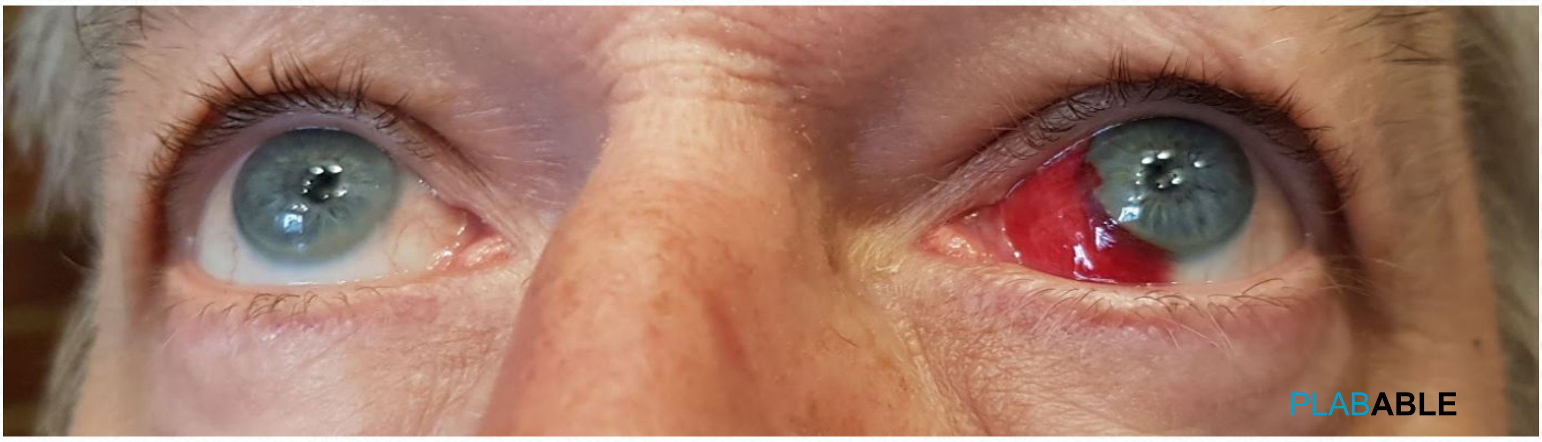
Retinitis Pigmentosa

Brain trainer:

Young adult patient who complains about night blindness followed which has worsened now to include peripheral visual field loss. Family history of a similar condition which has progressed to blindness.What is the diagnosis?

➔ **Retinitis pigmentosa**

Subconjunctival Haemorrhage



Brain trainer:

Above patient presents with a bloodshot eye. He denies pain, vision loss or ocular discharge. He is normotensive. What is the diagnosis and management ?

➔ **Subconjunctival haemorrhage**
➔ **Reassurance**

Herpes Zoster Ophthalmicus

Reactivation of the varicella zoster virus in the area supplied by the ophthalmic division of trigeminal

Features

- Preherpetic neuralgia
- Rash on the forehead and eyelid swelling
- Eye pain and photophobia
- **Hutchinson's sign:** cutaneous involvement of the tip of the nose (nasociliary nerve)

Treatment

- Oral antivirals - aciclovir
- Oral corticosteroids

Orbital Cellulitis

Presentation
- Proptosis
- Pain with movement of the eye
- Blurred vision
- Diplopia
- Swelling of conjunctiva and lids

Best initial test: CT scan of orbit and brain (rule out intracranial abscess)

Treatment
- Emergency referral to higher center
- IV antibiotics

Retinal Detachment

Predisposing factors

- Advancing age
- Extreme myopia
- Trauma and cataract surgery

Presentation

4 F

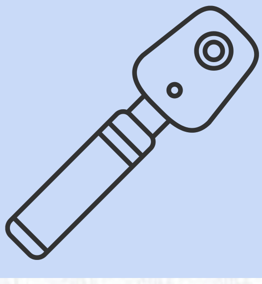
Flashes

Floaters

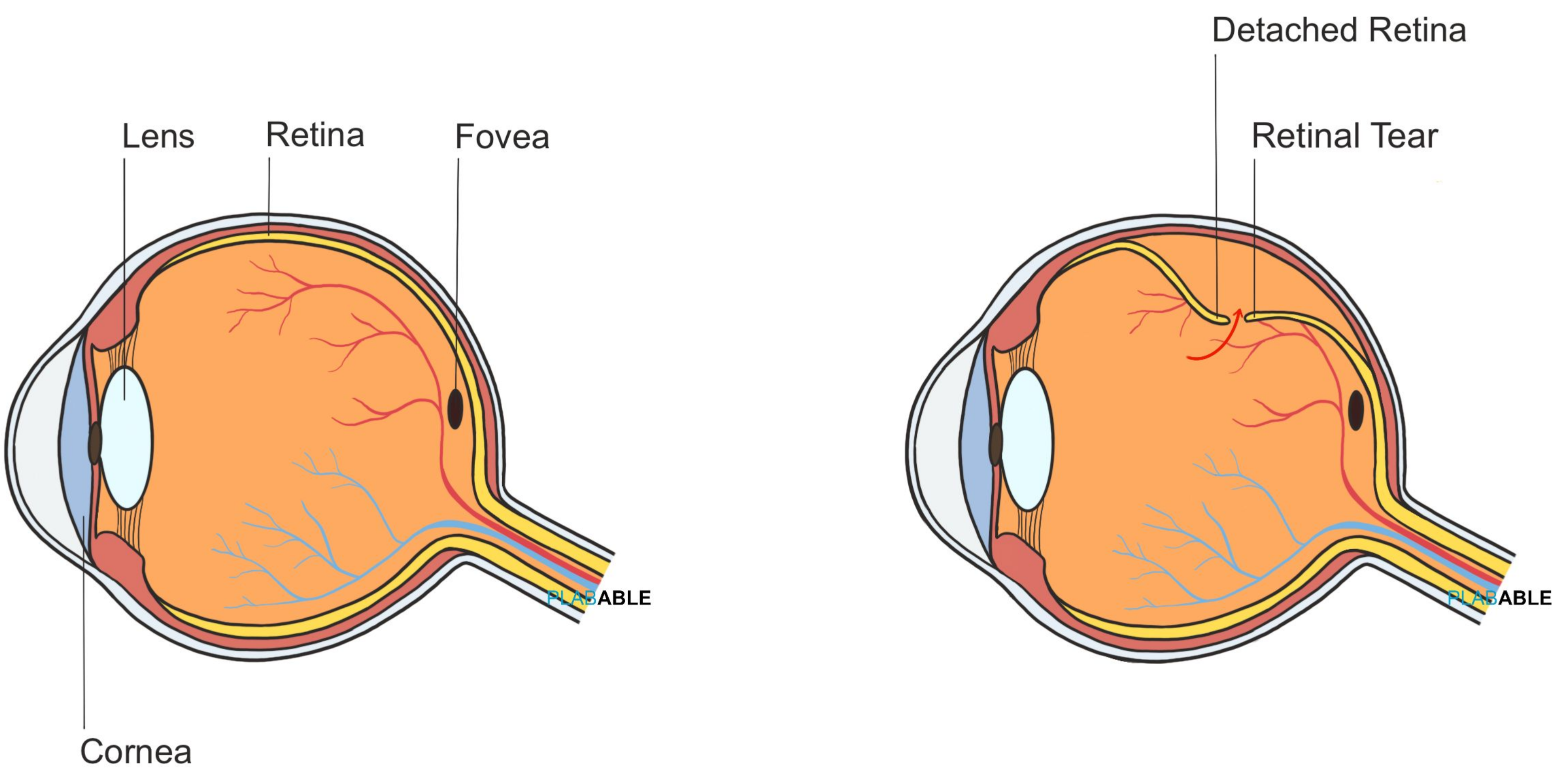
Field loss

Fall in acuity

Ophthalmoscopy: Grey, opaque and wrinkled retina that balloons forward



Retinal Detachment



Healthy Eye **Retinal Detachment**

Treatment

- Vitrectomy
- Scleral buckling
- Pneumatic retinopexy
- Retinal tear and holes are treated by cryotherapy or laser photocoagulation to prevent progression to RD

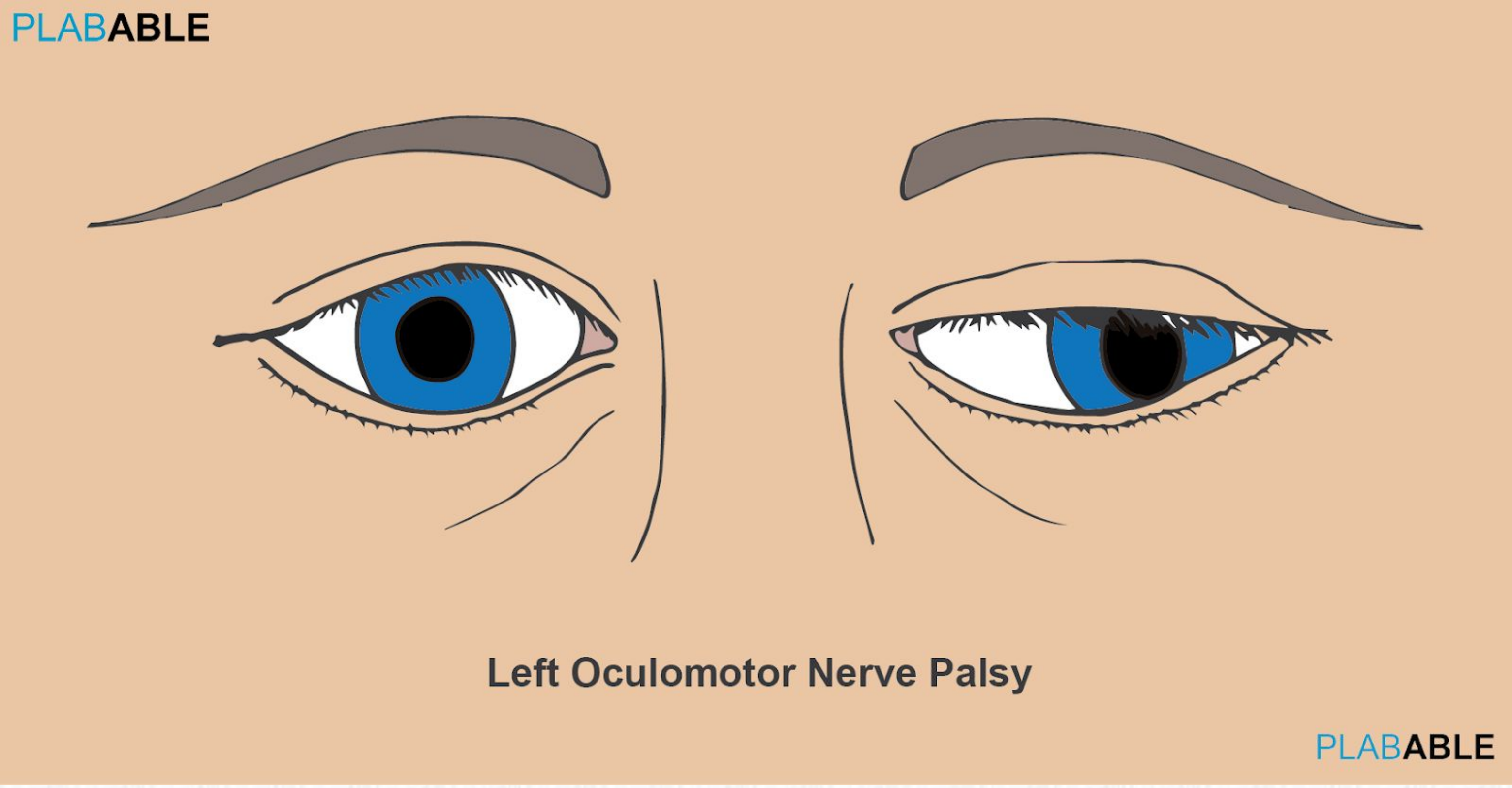
Oculomotor Nerve (CN3)

Function

- Pupil constriction (sphincter pupillae)
- Eyelid elevation (levator palpebrae superioris)
- Most muscles of eye movement

Mnemonic: SO4 LR6
Rest supplied by CN3

Palsy:



"Ptosis"

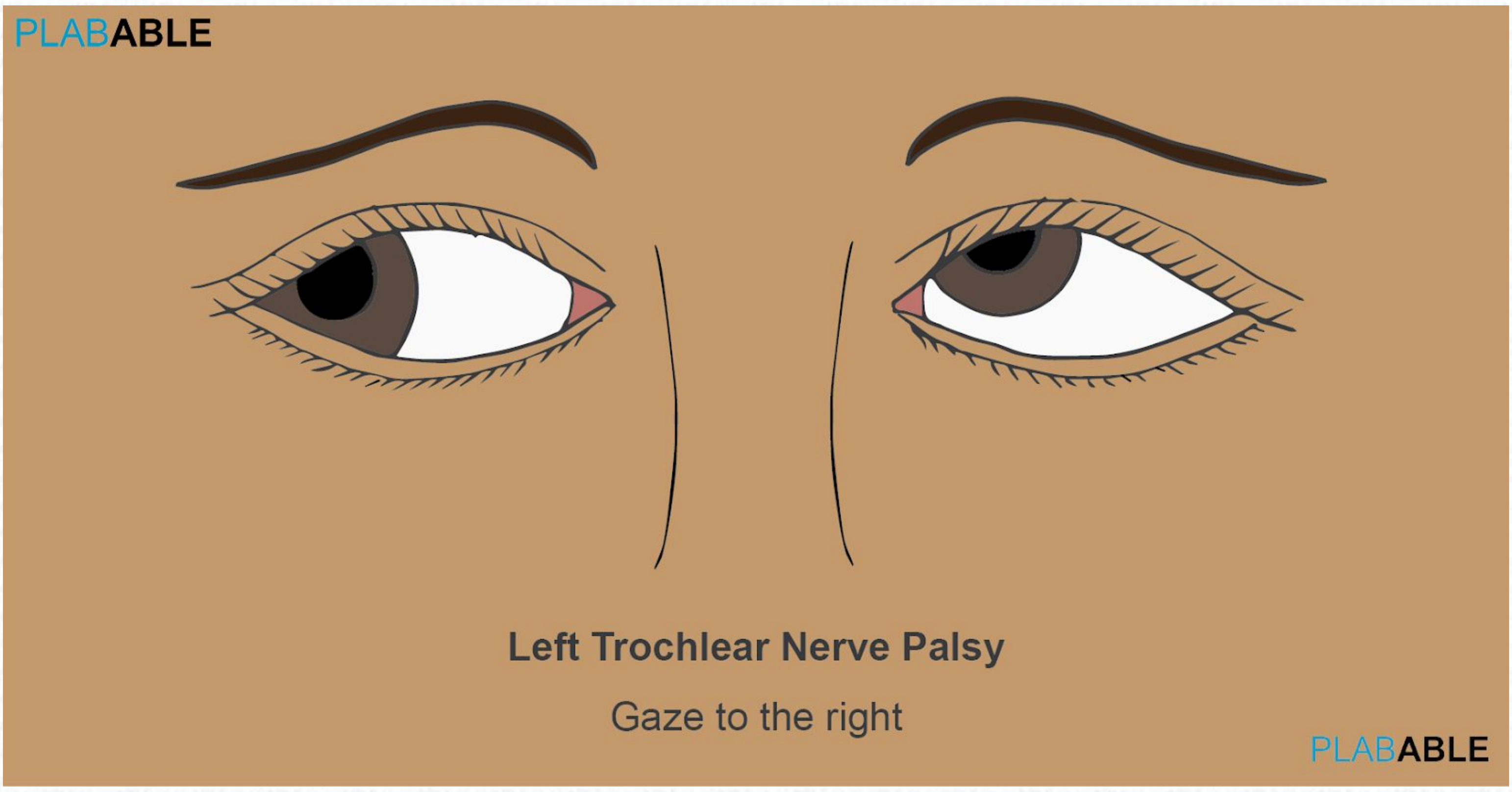
"Ocular dilation"

"Down and out"

Trochlear Nerve (CN4)

Function:
Inward and downward movement of the eye (superior oblique)

Palsy:



Patient (in picture)

1. Diplopia when looking to their right (i.e. opposite the lesion)
2. Tilt head to their right (i.e. opposite the lesion) to compensate.

- Reading
- Walking down stairs

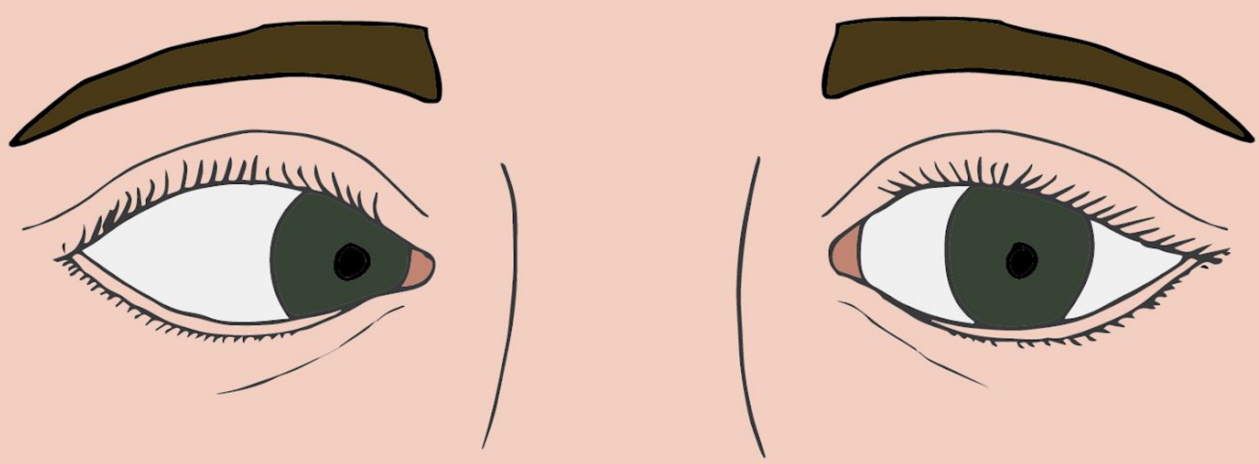
$\Big\}$ **Diplopia**

Abducens Nerve (CN6)

Function

Allows eye to look outwards (lateral rectus)

Palsy



Left Abducens Nerve Palsy

Gaze to the left

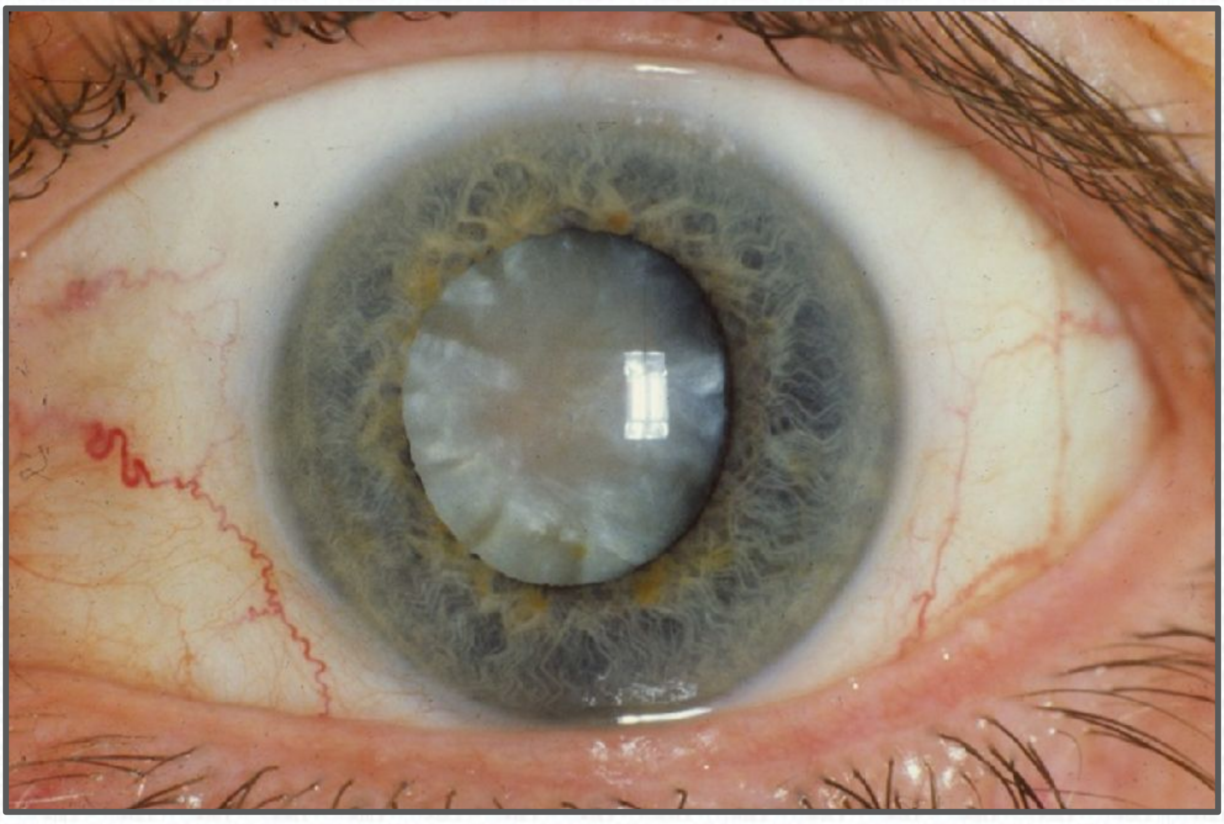
Patient (in picture) will have diplopia when looking to their left (i.e. same side as the lesion)

Cataract

Clues to diagnosis

- Old age
- Gradually decreased vision
- Glare especially at night
- Use of steroids (COPD or asthma)
- Increased exposure to UV (person from Australia)
- Diabetes and smoking are other risk factors

Treatment involves extracapsular lens extraction followed by intraocular lens implantation



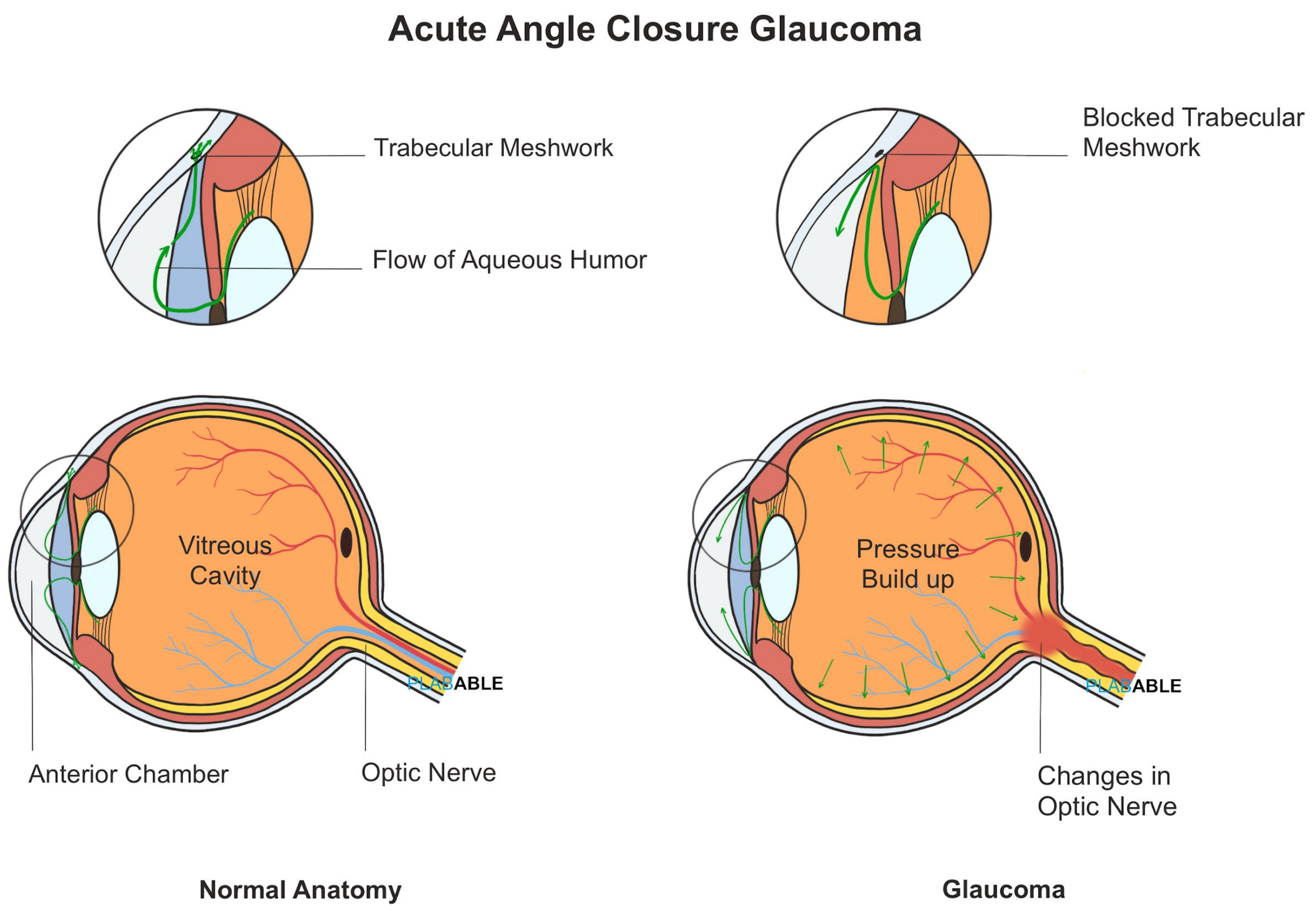
Note: Congenital cataract is mainly caused by **Rubella**

Acute Angle Closure Glaucoma

Increased IOP due to impaired aqueous outflow

Presentation

- Painful red eye
- Semi-dilated non-reactive pupil
- H/o coloured halos
- H/o watching TV in dark room (causes pupillary dilation and decreases outflow)
- Globe feels hard on palpation
- **Risk factor:** Hypermetropia



Acute Angle Closure Glaucoma

Acute treatment

- Intravenous acetazolamide
- Timolol drops (beta-blocker)
- Prednisolone drops
- Pilocarpine drops

Long term management

- Peripheral iridotomy
- Surgical iridectomy

Note: If the condition is not treated immediately, the increased IOP can damage the optic nerve and cause permanent vision loss

Primary Open-Angle Glaucoma

Progressive optic nerve damage due to ↑ IOP

Presentation

- Gradual visual field loss
- Usually bilateral
- Open iridocorneal angle
- Optic neuropathy - Disc cupping

Risk factors

Myopia, ocular hypertension and age > 65

Treatment

Medical:

- Prostaglandin analogues - First line (topical)
- Beta-blockers (topical)
- Acetazolamide

Laser and surgical:

- Laser and surgical trabeculoplasty
- Laser cycloablation
- Artificial shunts

Optic Neuritis

Inflammation of optic nerve and is seen in multiple sclerosis and neuromyelitis optica

TRIAD

Pain on eye movement

Rapid decrease in central vision (unilateral)

Impaired colour vision

Classically presents in the exam:

1. **Woman**
2. **Fatigue**
3. **Paresthesia**
4. **Weakness**
5. **RAPD**

$\left.\begin{array}{l}\text{1. Woman}\\ \text{2. Fatigue}\\ \text{3. Paresthesia}\\ \text{4. Weakness}\end{array}\right\}$ **Multiple sclerosis**

Treatment is with methylprednisolone

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Relative Afferent Pupillary Defect

No light

Normal response to light

Positive RAPD of Right Eye

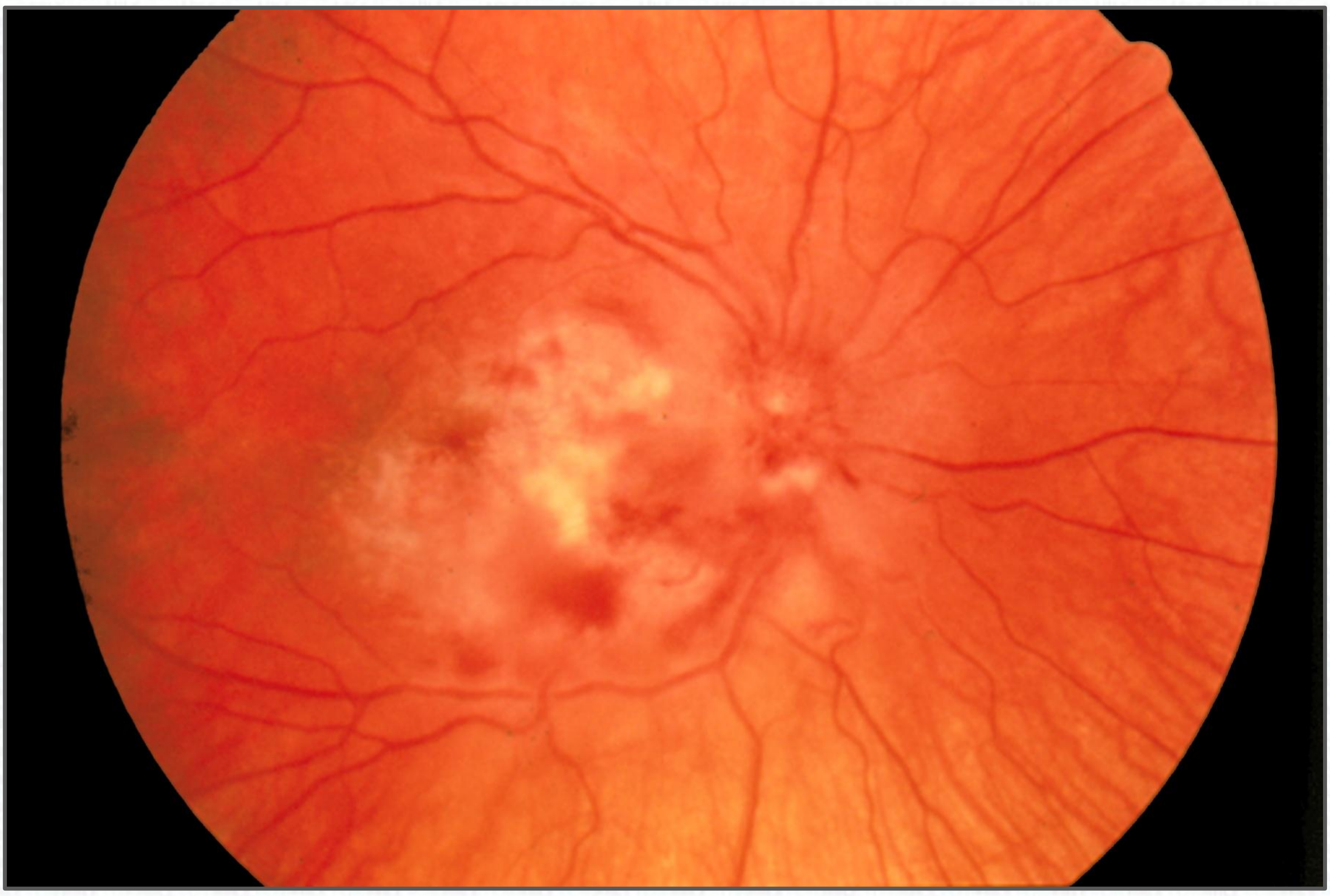
Cytomegalovirus Retinitis

Features

- **Immunocompromised** patient: AIDS, organ transplant or on chemotherapy
- Progressive visual deterioration
- Floaters and loss of visual field on one side
- **Fundoscopy** -
  - Retinal haemorrhages
  - Yellow-white areas (pizza appearance of retina)

Treatment

- Intravitreal and intravenous **ganciclovir**



Amaurosis Fugax

Ischemia

Vision loss

Sudden **Transient**

Painless

"Black Curtain coming down"

Associated with:

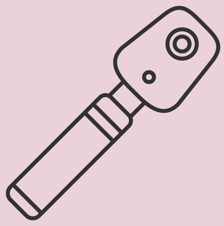
1. Transient ischemic attack
2. Giant cell arteritis

Central Retinal Artery Occlusion

Presentation

- **Acute painless loss of vision**
- H/o amaurosis fugax in the past
- Relative afferent pupillary defect

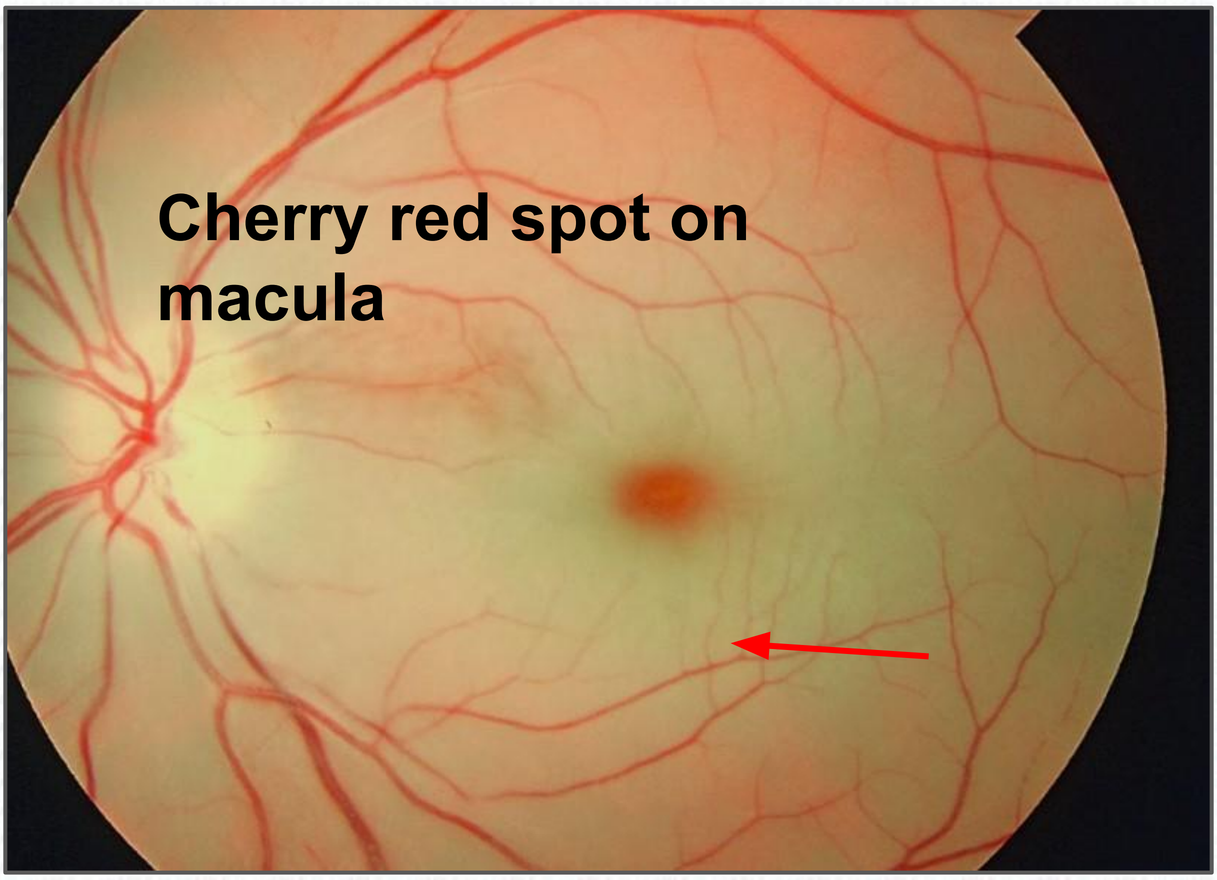
Ophthalmoscopy



- Pale retina
- Cherry red spot

Treatment

- If within 90 min of onset of symptoms then firm ocular massage to dislodge the clot
- Intra-arterial fibrinolysis



Central Retinal Vein Occlusion

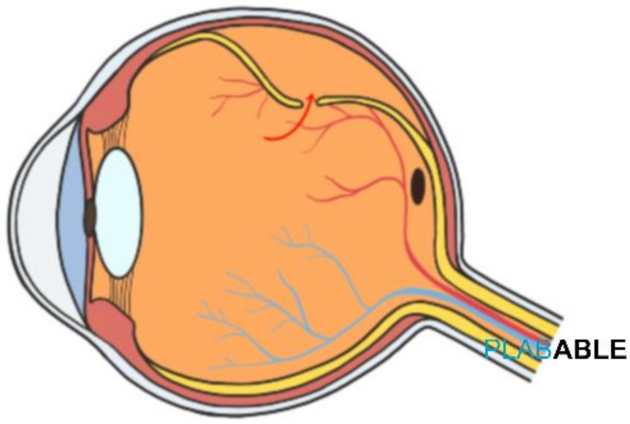
Presentation
- Unilateral, painless loss of vision or blurred vision
- Image distortion
- Field defect

Fundoscopy
- Dot-blot and flame haemorrhages
- Macular oedema

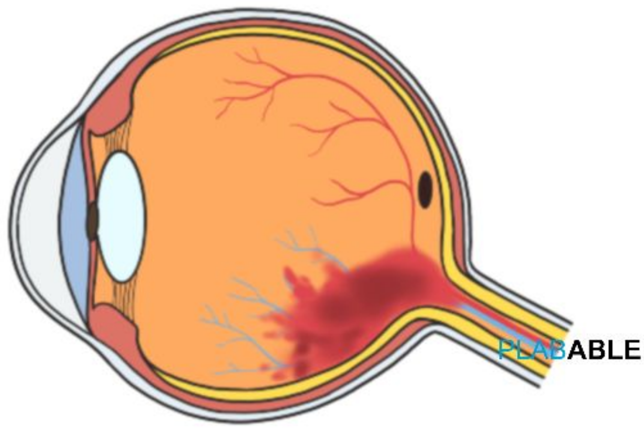
Treatment
- Panretinal photocoagulation
- Intravitreal anti-VEGF

Causes of Sudden Painless Vision Loss

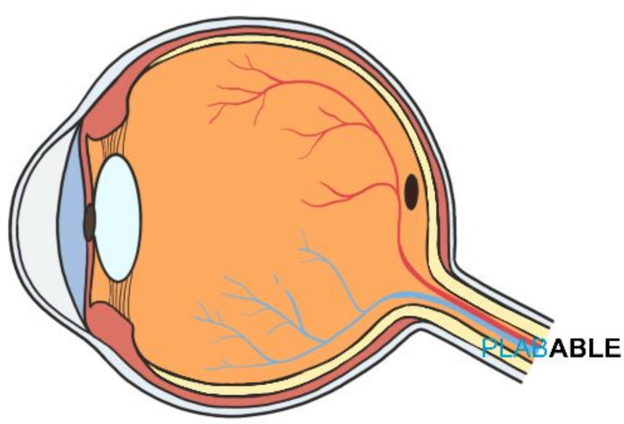
1. Retinal Detachment



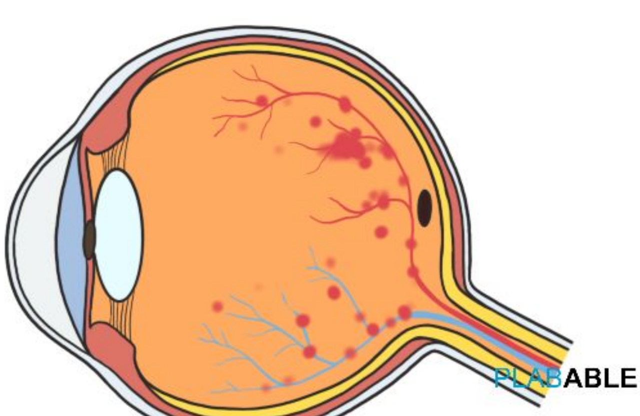
2. Virtuous Haemorrhage



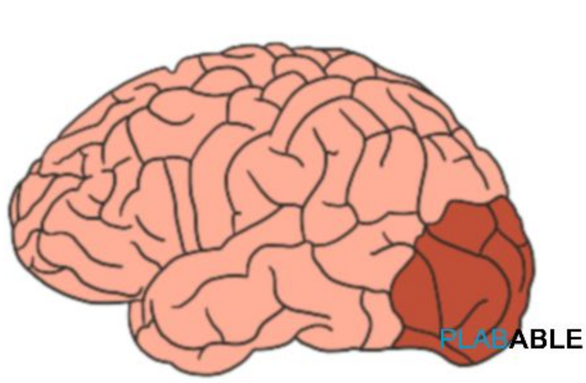
3. Central Retinal Artery Occlusion



4. Central Retinal Vein Occlusion



5. Occipital Cortex Stroke



Diabetic Retinopathy

Nonproliferative or background retinopathy

- Microaneurysms (dots)
- Haemorrhage (blots)
- Hard exudates

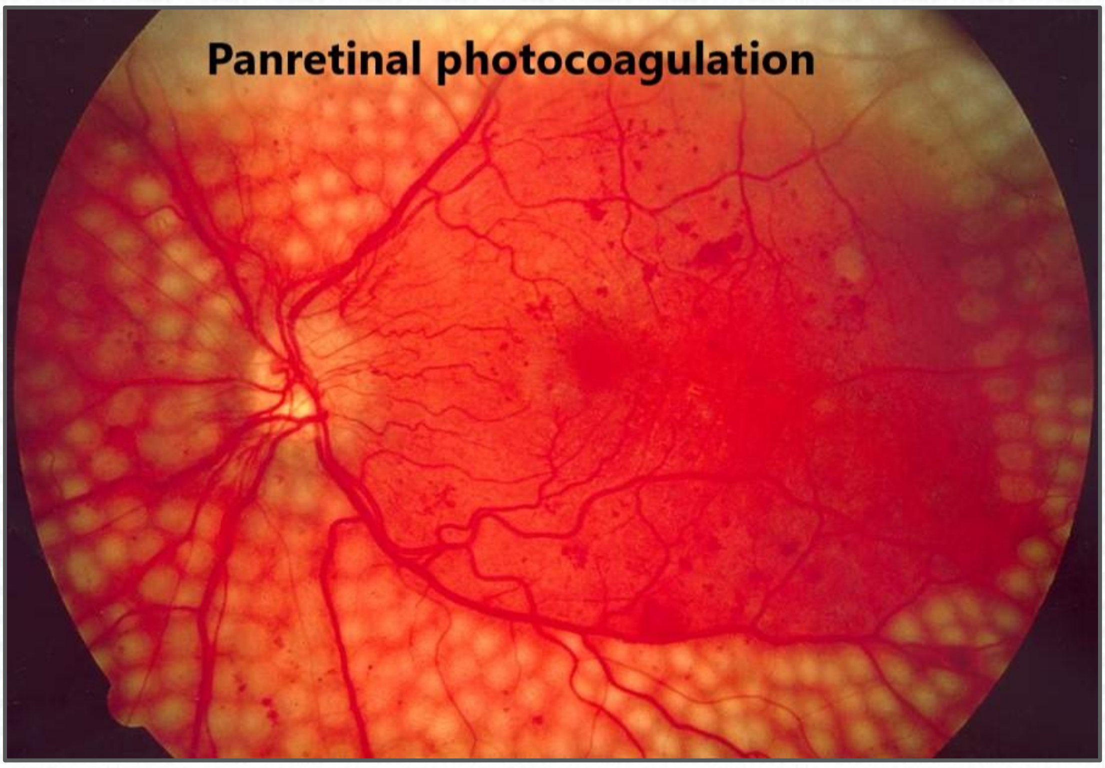
Preproliferative retinopathy

- Addition of cotton wool spots

Proliferative retinopathy

- Addition of neovascularization
- Vitreous haemorrhage
- Floaters

Treatment: Laser photocoagulation, intravitreal steroids, and anti-VEGF



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Hypertensive Retinopathy

Presentation:

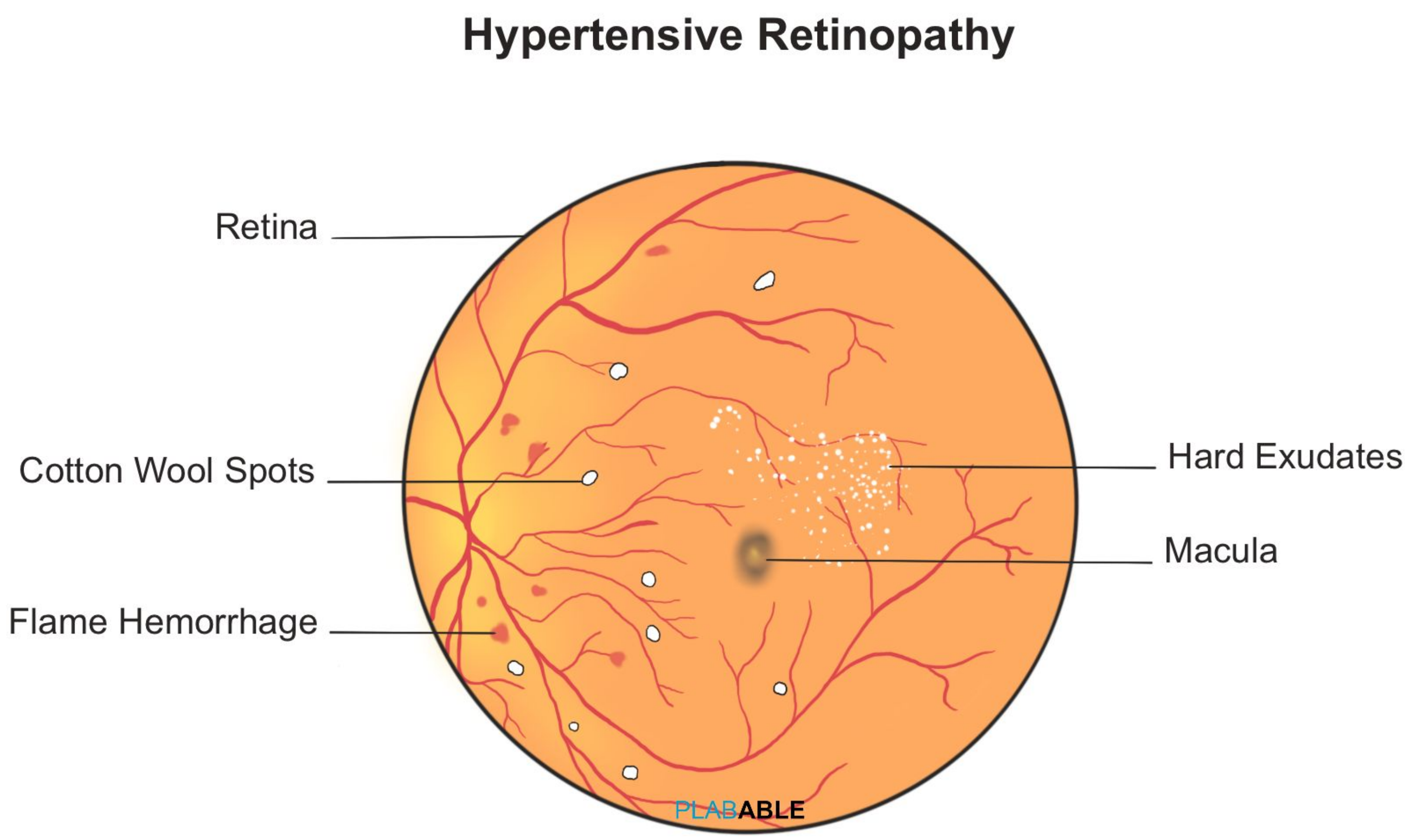
- **History of uncontrolled hypertension**

Fundoscopy:

- **AV nicking** (where an artery crosses a vein)
- **Copper or silver wiring** (attenuation of artery)
- **Cotton wool spots**
- **Flame shaped hemorrhage**
- **Optic disk oedema and ischemic changes**

Management:

- **Treat hypertension**



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